

# Config Drift Detector - Project Explanation

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## 1. Project Overview

The Config Drift Detector is an AI-powered system designed to analyze, track, and explain discrepancies (drifts) between an intended configuration profile and the actual, live deployed configuration. In modern cloud operations and DevOps, configurations are prone to manual adjustments, software updates, or unauthorized modifications. This project provides a user-friendly dashboard and automated pipeline to keep configurations aligned and secure.

## 2. Technology Stack

The system is split into a modern decoupled architecture:

- Frontend:** React (built with Vite) for a lightning-fast development experience and optimized bundle. Stylings are structured using modern dark mode custom CSS variables. Charting is powered by Recharts (responsive Pie, Bar, and Area charts), icons by Lucide-React, and routing via React Router DOM.
- Backend:** FastAPI (Python) serves as the REST API engine. It is designed to handle high-throughput file uploads asynchronously. Uvicorn acts as the ASGI server wrapper.
- AI Core:** Google GenAI (Gemini) is integrated to analyze detected configuration changes, explain their potential impacts, and generate contextual recommendations on how to remediate the drift.
- Analysis:** DeepDiff is utilized to perform deep recursive comparison of objects, keys, lists, and values regardless of order or format in JSON and YAML configurations.

## 3. Core Functionalities

### - Real-time Dashboard

Aggregates key system metrics including total analyzed files, breaking issues, functional issues, and cosmetic issues. Includes responsive visual charts (Severity Distribution Pie Chart, Drifts Per Report Stacked Bar, and Cumulative Drift Trend Area Chart) and a Recent Activity feed that automatically pulls from history.

### - Drift Analysis Page

Allows users to upload or drag-and-drop 'Intended' and 'Actual' JSON/YAML configuration files. Processes them through the backend comparison engine and lists detailed drift key breakdowns alongside AI-generated remediation recommendations.

### - Interactive Reports History

Saves all previous analyses locally in localStorage. Users can filter, search, view full historical comparisons, delete specific records, or download them as Markdown or PDF files.

### - Comprehensive Settings

A settings center containing configuration for themes (dark, light, system modes),

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customizable UI accent colors, toggleable notifications, severity threshold filtering, storage limits, and API server endpoint overrides.

### 4. Data & Analysis Pipeline Flow

1. File Upload: The user uploads intended and actual JSON/YAML files via the Drift Analysis tab.
2. Diff Calculation: FastAPI reads the files and uses DeepDiff to analyze additions, removals, and changes.
3. Severity Classification: The backend categorizes changes into Breaking, Functional, or Cosmetic.
4. AI Augmentation: The diff content is dispatched to Gemini to produce plain-english impact analysis.
5. Dashboard updates: React updates localStorage which pushes real-time updates to the active widgets.